Reviewed mixed-degree addition components

Cancellation: field and word

Input validity implies ordinate +/- alternatives, excludes mismatch branch; cancellation output has nonzero weights and normalized validity. Field; shared support; both normalized input divisibilities; four nonzero weights. Repeated root/zero ordinate allowed.

Exact prelude/degree-one output decode and canonicality; guards agree; conditional word cancellation validity. Prime 5<=p<2^31; 10 canonical coordinates; four nonzero weights; both normalized input validities.

Generic branch: field

Interpolation and explicit degree-two residual establish nonzero output weights and normalized validity under generic K!=0. Field; monic quintic without X4; four nonzero weights; both input validities; K!=0. Repeated output support and zero interpolation slope are allowed. This establishes polynomial divisibility, not a divisor-class sum.

Generic branch: literal UInt64

Six canonical decoded payloads; zero guard and Option correspondence; outer/generic K equality; conditional output validity. Prime 5<=p<2^31; 10 canonical coordinates plus curve[3]; validity adds nonzero weights, input validity and K!=0. The actual generic formula uses explicit squares; no pow loop. Literal output w=1 is canonical. The other curve coefficients enter the validity statement as field values.

What the reviews checked

Separate module, control and axiom readbacks passed. Cancellation review covers both field and word modules; generic field and generic word have separate reviews. Only standard Lean axioms were read back. Original failed attempts and control corrections remain in their packets.

Source assumptions and remaining branches

Machine scope is Field n=1, tables=None and normal counter execution. Pure value bundles may define unused outputs beyond an early-return guard; no operation-count or side-effect equivalence is asserted. No compiler/linkage, class-sum, tripling, secondary positive-coincidence output or whole-walk theorem is claimed. Polynomial controls do not assert smooth curve preprocessing.

Publication and software snapshot

Canonical integration is pending final root configuration. Boxed maintained Clippy: exit0, sources unchanged. Typed retained-data tests, full library and CLI checks remain pending root terminal review in this snapshot. These software gates are separate from mathematical component review.